6

HACCP implementation

The Indian experience

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6.1 Introduction

In recent years, the hazard analysis critical control point (HACCP) system has become the accepted safety benchmark for food systems. The HACCP system is a management technique that allows a company to examine systematically all the processing steps involved in manufacture and use of a product, and then identify and manage those processes/operations that are critical to the safety of the product. HACCP systems have become mandatory for a wide range of foods in many countries and are now required for any exports out of India. In this chapter, issues related to HACCP development, implementation and acceptance in India are discussed. Given the disorganised nature of most of India's food industry, establishing effective systems for assuring food safety is an uphill task. There are also serious challenges in developing and sustaining work in areas such as disease surveillance and food safety research that may hamper the effective implementation of HACCP in India.

6.2 The food industry in India

India is the world's second largest producer of foods, and it has been estimated that food production will double from current levels by the year 2005. A quick survey of India's food production reveals the magnitude and the potential of the industry:

- India is the largest producer of fruits and the second largest producer of vegetables in the world.

- Indian dairy production is 70 million metric tonnes (MT), the second largest in the world.
- India produces and exports all the main varieties of spices. Spice production is worth over Rs 35 billion (approx. $800 million) and accounts for 25–30% of world production.
- India grows 22 MT of oilseeds annually, covering most of the varieties.
- Important plantation products from India include tea, coffee, cocoa and cashew.
- India has a large marine products and processing potential with varied fish resources along the 7,500 km long coastline, 28,000 km of rivers and millions of hectares of reservoirs and lakes.
- India's livestock population is the largest in the world with 50% of the world's buffaloes and 20% of cattle, but only a very small percentage (1%) is converted to value-added products.

The scale of overall Indian food production means that the processed food industry will become a very important part of the Indian economy. Consumption of value-added foods is estimated at approximately Rs 225,000 ($52 billion) per annum. The contribution of food processing and production to India's GDP is about 26%. The size of the semi-processed and ready-to-eat packaged food sector is Rs 40 billion (approx. $1 billion) plus per annum and growing at over 20% year on year. Increased urbanisation has also allowed the soft-drink market in India to grow rapidly, and is currently worth Rs 17–18 billion (approx. $500 million).

The overall size of the food industry is, however, not matched by its efficiency. Antiquated practices ensure that India's agricultural production is well below world averages. As an example, wheat and rice yields are 32% and 33% of world levels respectively. These practices are coupled with inefficient storage and distribution, which have led to tremendous wastage. It is estimated that about 40% of the total production of fruits and vegetables is wasted in the distribution chain. This wastage is due to the following:

- variable crop quality
- poor handling and storage on the farm
- an inadequate transport system
- too many intermediaries in the supply chain
- poor storage facilities.

6.3 Food safety in India

India has all the problems associated with a developing country, namely a large population, rapid urbanisation, widespread poverty and lack of access to basic sanitation and hygiene for much of its population. Foodborne illness statistics are not very easily obtainable or do not exist. Diarrhoeal diseases are very common throughout the country. On occasions when there are natural calamities

and the existing infrastructure breaks down, there are severe epidemics of infectious diseases, for example cholera and gastroenteritis. However, the contribution of foods to the overall incidence of disease is not known. In studies that have been carried out at the Hindustan Lever Research Centre (HLRC), fresh vegetables were found to contain coliforms at levels ranging from 1×102 to 4×104 Colony Forming Units per gram (CFU/g) (unpublished data). Although comprehensive data are lacking, the situation with meat products is likely to be worse, given the following:

- the lack of good animal husbandry practices in breeding and rearing
- the poor resulting quality of the animals and poultry used for meat
- very poor hygiene in the slaughterhouses where meat is processed
- inadequate systems for storage, distribution and display of meat for sale.

There is also little control over the quality of drinking and process water. The HLRC has carried out studies that show that potable water across the country does not conform to WHO guidelines for safety and quality (unpublished data). When such water is used for food processing, the risk of microbiological contamination is increased significantly.

India, like other developing countries, does not have the necessary infrastructure to maintain the safety of perishable food products. Frequently transport and warehouse conditions are such that food products spend a very long time under undesirable conditions of high temperature, high humidity and poor hygiene. The cold chain in India is very poorly developed, and temperature abuse during storage and distribution can lead to serious safety defects in perishable refrigerated or frozen products such as ice cream.

As well as microbiological contamination, the food industry in India faces significant chemical hazards. Pesticides, for example, are widely and sometimes indiscriminately used with a resulting risk to health. The Indian Council of Medical Research (ICMR) and the Directorate General of Health Services (DGHS), in collaboration with the WHO and the Food and Agriculture Organisation (FAO), have conducted several surveillance studies in India on pesticide residues in fruits and vegetables, which have shown both the use of illegal pesticides and the excessive use of those that are permitted. These pesticides and their residues have also been shown to occur commonly in waters affected by agricultural discharges and make their way into potable water supply.

The food industry in India has to contend, therefore, with major problems in the supply chain. However, its difficulties are not just material. They are also cultural. In a country where overpopulation and lack of food dominate, food safety has not traditionally been a priority. While epidemiology and surveillance mechanisms are poor, the public perception of the risks of foodborne illness is also very low. Consumers do not believe that foodborne illness is a major issue in India, resulting in no strong consumer lobby for food safety. Thus it is difficult for the food industry to sell the concept of food safety and HACCP systems to consumers. This attitude has reduced the

commercial rationale for improving food safety in relation to the costs of making such improvements.

In India the food processing industry has been a relatively small sector of the economy, and has only begun to grow in the last decade. In addition, until the liberalisation of the economy in the early 1990s, the Indian economy was largely inward-looking and little influenced by external competition. Until recently, there was little systematic quality management or even agreed standards of good manufacturing practice (GMP). The food processing industry's initial focus was on product and market development, reducing further the impetus to improve food safety systems. Hygiene and safety were part of a cycle where legislation was not very effective and there was very poor surveillance of foodborne disease, which led to an underestimation of risk, which in turn did not provide incentives for the food industry to implement better food safety systems.

Indian food laws are also part of the problem. There are more than 20 different pieces of legislation governing foods which are difficult to follow or regulate. Most of the legislation, for example the Prevention of Food Adulteration (PFA) Act, is very prescriptive and punitive. The PFA Act, for example, was developed in the 1950s when food adulteration was a very serious issue. In recent years, where the emphasis is on safety and quality, such legislation has lost most of its relevance and needs to be rewritten. A committee appointed by the government of India has indeed made the recommendation that a new Food Safety Act be created.

6.4 The development of HACCP systems in India

The slow take-up of HACCP principles in India only began to change with the liberalisation of the Indian economy in the early 1990s. This process had two effects, one of which was to expose Indian food manufacturers to significant foreign competition with far higher standards of product safety and quality. It also stimulated the growth of Indian food exports, and it is in this sector that HACCP implementation has subsequently made most progress, largely in response to the strict requirements on safety and quality enforced by major trading partners such as the USA and EU. As a result of early consignments being blocked as unsafe, the seafood export industry in particular has undergone fundamental changes in the way that it tackles hygiene and safety matters. HACCP systems are now mandatory for this industry. A similar situation prevails in the highly lucrative area of export of fresh fruits.

Given the importance of exports to India's economy, the government has recently proposed several new measures and incentives to ensure safe food processing. The Agricultural Produce Export Development Authority (APEDA), a government agency, provides training and guidance on a regular basis on HACCP implementation and related issues. The APEDA also provides fiscal incentives to manufacturers to implement HACCP in their plants. The majority

of exporters are small manufacturers and such incentives are powerful motivators for change. The Bureau of Indian Standards (BIS), the principal institution setting product and process standards for manufacture of all types of products, has recently launched a HACCP certification scheme for food manufacturers. Other bodies introduced by the government to promote and enforce HACCP systems in the export sector include the Export Inspection Agency (EIA) which audits and certifies HACCP systems against international standards; the Central Institute for Fisheries Education (CIFE), specifically in the marine products sector, which provides training support; and the Marine Products Exports Development Agency (MPEDA), which provides advice to exporters on international standards.

There has been a major change internationally in attitudes toward food safety as a result of the deliberations at the Uruguay Round of the General Agreement on Tariffs and Trade (GATT) talks. Due to the implementation of the World Trade Organisation (WTO) rules, food safety procedures are now to be governed by the WTO sanitary and phytosanitary (SPS) agreement. The Codex Alimentarius is now the internationally recognised body for implementing food safety systems. India, as a signatory to GATT, needs to ensure that both the domestic and export sectors of its food industry conform to HACCP principles. It is therefore a good sign that India is an active participant in the deliberations of the Codex Alimentarius Committee on Food Hygiene. India has emerged as the voice of the developing countries in these discussions.

6.5 HACCP implementation: the Unilever approach in India

Globalisation of food trade has allowed a number of important new international players to enter the Indian market. Among these are multinationals such as General Foods, McDonald's, Coca-Cola, Pepsi Cola and Unilever (through its Indian arm, Hindustan Lever Limited (HLL)). In addition, companies like Godrej and other Indian multinationals have significantly strengthened their portfolios in foods. A key point of differentiation between these companies will be the quality and safety of the foods they produce.

Among the various food manufacturers in India, HLL has emerged as the leader in developing and implementing food safety systems over the last eight to ten years. Unilever's food safety systems world-wide are based on the principle of 'safety by design'. This concept is based on prevention. Its key features (illustrated in Fig. 6.1) are as follows:

- product formulations are designed to be inherently safe
- processes are designed in such a way that they deliver safe products.

It is important that these two elements work closely together, since not even the safest product design will deliver a safe product if manufacturing processes are not safe. 'Safety by design' is essentially a farm-to-fork approach that allows Unilever to ensure safety throughout the manufacturing and distribution chain.

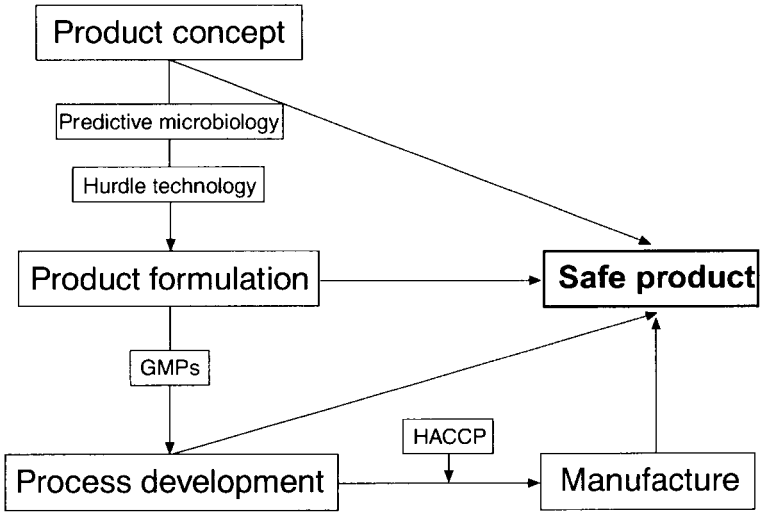


Fig. 6.1 The Unilever philosophy of food safety.

This concept relies on the existence of a variety of robust systems that ensure sustained delivery of safe products. These systems include the following:

- robust, safe product design tools and information (e.g. extensive micro-biological research facilities providing skills in predictive microbiology and the understanding of interactions between micro-organisms and product components)
- GMPs and allied systems for cleaning and sanitation to ensure safe manufacture; also included are minimum standards for animal husbandry and agricultural practices, and systems for vendor and supplier approval
- continuous training for all personnel involved in food formulation design and manufacture to ensure that food safety is an ingrained habit
- robust HACCP systems for safe manufacture.

These are supported by the availability of uniform analytical methods across the company, establishment of formal quality assurance (QA) systems, and a structured, comprehensive system of auditing and review. Within HLL, the concept of 'safety by design' has been successfully established in food manufacture and is being extended to non-food products.

6.6 HACCP implementation in practice: the marine products model

HLL exports a variety of marine products to Japan, the USA and the EU. These products include frozen shrimp, cuttlefish, fish fillets, lobster and surimi (frozen fish paste). These products are generally reprocessed by the customer into final

products for sale to the consumer. The marine products business has a current annual turnover of approximately Rs 700 million ($16 million) and employs 1,000 people at six manufacturing locations. In 1992, it was decided to implement HACCP systems in marine products manufacture, much ahead of proposed US FDA and EU regulations for seafood imports. As a result of this initiative, the HLL marine products business is now fully HACCP-compliant. HLL has the largest surimi manufacturing facility in India and this was the first such unit to have a fully implemented HACCP system.

6.6.1 Implementing HACCP: the main phases

The Microbiology Section at the HLRC was assigned the task of planning and implementing HACCP systems. HACCP implementation was planned in a series of phases. A preliminary evaluation was carried out to determine the state of readiness of each site. Based on this initial evaluation, the HACCP implementation process was divided into the following phases:

Phase 1: an initial awareness and training programme
Phase 2: a comprehensive HACCP training programme
Phase 3: preliminary audits of all processing units
Phase 4: training of analytical personnel
Phase 5: HACCP implementation itself
Phase 6: certification of HACCP systems.

Overall, the first five phases took 18 months to complete.

6.6.2 Phase 1: the initial awareness and training programme

While the marine products division had an existing QA-based system for manufacture, it relied largely for quality control on sampling and inspection of the final product. Although this fitted in with customer requirements, it was not effective in identifying and resolving safety problems during the production process itself. This reflected wider weaknesses in management. Knowledge of food safety issues and microbiological hazards was very limited in the organisation at that time. Phase 1 therefore involved an initial training and awareness programme for senior managers. The underlying theme of training was that food safety is both a technical and a management issue and, to be most effective, has to be tackled through an appropriate management system such as a HACCP system. It was also emphasised that commitment to food safety has to be top-down and should be demonstrated at all levels within the organisation.

In the first phase of training, business and technical heads participated, along with factory management. About 20 people were trained in this initial phase. Training was carried out by staff from HLRC as well as by experts from the CIFE. Subject areas covered included:

- principles of food safety and the concept of safety by design
- the philosophy of quality assurance
- management commitment to food safety
- GMPs and good hygienic practices (GHPs)
- food microbiology, particularly foodborne pathogens
- principles of sampling and analytical methodology.

The course was successful in winning the commitment of senior management to the HACCP programme and agreeing a schedule for the remaining phases. The main issue arising out of the training course was the need to upgrade prerequisite systems, especially at third-party owned co-packers.

6.6.3 Phase 2: the HACCP training programme

In phase 2 a comprehensive HACCP training programme was conducted for functional personnel. These personnel were selected to form the HACCP implementation teams for the respective factories. These teams included people from the production, supervisory, quality assurance, distribution and hygiene functions. All team members were selected on the basis of their expertise and included, as far as possible, people who were closely involved with day-to-day production operations. Due to the prevailing level of education of operators and workers, it was necessary to have only supervisory personnel in the core HACCP teams.

Staff from the Microbiology Section at HLRC carried out this training. The training module covered the following:

- HACCP principles and the benefits of HACCP systems
- detailed implementation procedures
- case studies illustrating HACCP implementation in practice.

The teams proved extremely enthusiastic about the benefits of HACCP and were charged with the responsibility of cascading down the message of HACCP to shop-floor personnel, and to develop the appropriate training modules for line operators and workers.

6.6.4 Phase 3: preliminary audits of all processing units

In phase 3, preliminary audits of all processing units were conducted to gauge their state of readiness for HACCP implementation. Each factory was audited against internal Unilever standards for safe manufacture and those set by the MPEDA, a government body, and the EIA, the government's export audit and certifying agency. Where appropriate, individual buyer requirements were also used in the auditing process, particularly for larger customers.

The result was to identify a wide variation in the quality of prerequisite systems. A programme of improvements was agreed to rectify these and establish, as far as possible, uniform standards and systems across all factories.

Key improvements in prerequisite systems included:

- improved control of staff access to the production line to prevent, for example, the risk of cross-contamination
- modifying the production line layout to ensure a logical raw material/product flow
- redesign of areas and procedures for receipt and handling of raw materials
- the creation of designated high hygiene areas and procedures
- modifying anterooms to cold stores to ensure reduced levels of environmental contamination
- establishment of microbiology testing facilities with trained analytical staff.

A programme of improvements was agreed together with a schedule for completion. At the same time preliminary HACCP planning began with the preparation of draft process flow diagrams. At this point there was no attempt to carry out hazard analysis and CCP determination, given the need to complete the programme of improvements to prerequisite systems. A major training need identified at this stage was the training of chemists in microbiological analysis, since most factories did not employ microbiologists.

6.6.5 Phase 4: training of analytical personnel

Training of staff in microbiological analytical techniques was scheduled very soon after the preliminary audits of prerequisite systems. The purpose of this programme was as follows:

- to train analytical personnel in microbiological and analytical techniques for the purposes of CCP monitoring as well as local certification needs
- to enable them to subsequently set up a fully fledged analytical facility in the factory.

Factory analysts from the quality assurance laboratories, who would eventually oversee the monitoring of CCPs and other analytical testing requirements, were the target of this training programme. This was a comprehensive, hands-on training programme lasting ten days and covered all areas of microbiological analysis. The training programme contained the following modules:

- principles of microbial nutrition and growth
- sampling plans
- hands-on microbiological analysis
- certification.

The effectiveness of the training programme was gauged through feedback from trainees and a formal examination at the end of the course. The effectiveness of factory microbiological analytical facilities has subsequently been monitored during routine audits carried out in the factories. The MPEDA also routinely audits analytical systems and certifies analysts.

6.6.6 Phase 5: HACCP implementation

HACCP implementation was begun once the improvements to prerequisite systems identified in phase 3 were completed. The first part of the HACCP implementation phase involved the following at each factory:

- finalising and validating process flow charts
- the identification of hazards and CCPs
- preparation of the final documentation for HACCP plans, particularly CCP monitoring and record-keeping documents
- verification of HACCP plans at each of the factory sites.

During this process the HLR microbiology team took on the role of central consultants and facilitators, providing advice and disseminating best practice from one factory to another. At all times, however, it was emphasised that ownership of HACCP systems rested solely with the factories.

Once this stage was completed, one factory was chosen for pilot implementation. The choice of the factory was dependent on factors such as size of operations, level of preparedness and immediacy of need. Process flow charts were finalised and validated, in part to check that modifications agreed in phase 3 had been carried out. The HLR microbiology team spent two to three days at the factory undertaking a comprehensive review of the HACCP plan. Particular attention was given to the quality of hazard analysis, given the previous inexperience of factory staff in this area. Data used to review hazard analysis included:

- microbiological data from the factory
- information from published literature, government and other agency surveys
- the opinions of microbiology and manufacturing experts within the industry.

At the end of this process, the factory HACCP team took over responsibility for implementation. It was agreed that pilot implementation be carried out for a period of six months to work out all the wrinkles in the system and ensure its smooth functioning. After three months and at the end of the six-month trial, the central HLR team reviewed the effectiveness of the HACCP plan. The main problems encountered were with proper documentation of CCP monitoring data. In order to ensure more effective documentation, generic CCP monitoring data sheets were created for use by shop-floor personnel.

Halfway through the pilot study, implementation at the other units was started. Cross-factory visits were very useful for ‘live’ hands-on experience as well as knowledge sharing. At the end of the six-month period, HACCP was fully and successfully implemented at the pilot site and well advanced at the other sites. Since the manufacturing processes as well as the product portfolio were very similar across all the factories, completion of HACCP implementation at other sites was able to proceed quite smoothly. Within 18 months of the initial training exercise in phase 1, HACCP implementation was completed at all marine products units.

6.6.7 Certification

With the completion of phase 5, HACCP plans were submitted for certification audits by the EIA. The EIA bases its requirements for safe manufacture on a combination of EU as well as US FDA requirements. The EIA is also the approved certifying agency for HACCP in India. All HACCP plans were found to be compliant in audits carried out by EIA auditors. Since during phases 3 to 5 care was taken to ensure that factory operations were in conformance with all statutory requirements, only minor modifications were required after the EIA audits, confirming the success of the HACCP programme.

6.7 Maintaining HACCP systems

Effective functioning of HACCP systems requires continuous review and training to ensure that they remain effective and relevant. Factories have a standard HACCP review programme at least every six months. In addition, several other events will trigger an immediate review:

- changes in raw material, product, process
- any organisational changes affecting staff operating the HACCP system
- infrastructural or equipment changes
- newly published information on hazards that may affect the product
- changes in statutory or Unilever requirements.

All planned reviews, routine or otherwise, are documented. The HLRC central team continues to function as a central scientific resource to advise, for example, on emerging hazards. It also carries out training of new personnel or retraining of existing personnel whenever required, and routine annual audits to ensure the effective working of HACCP.

6.8 The benefits of HACCP implementation

Successful HACCP implementation has had several benefits. HACCP implementation has allowed factories to achieve better product safety and quality, reduce costs, design better products and processes, increase quality awareness among suppliers and improve team spirit. Based on interviews and on data collected as part of HACCP implementation, the following is a list of key benefits:

- HACCP implementation has led to an overall increase in productivity and production efficiency (Figs 6.2 and 6.3). As an example, in a typical ice cream factory in HLL, productivity went up from 3.75 Mh/Liton before HACCP implementation to about 5.5 Mh/Liton after HACCP implementation. This increase in productivity was the result of fewer quality defects interrupting production as well as some of the improvements listed below.

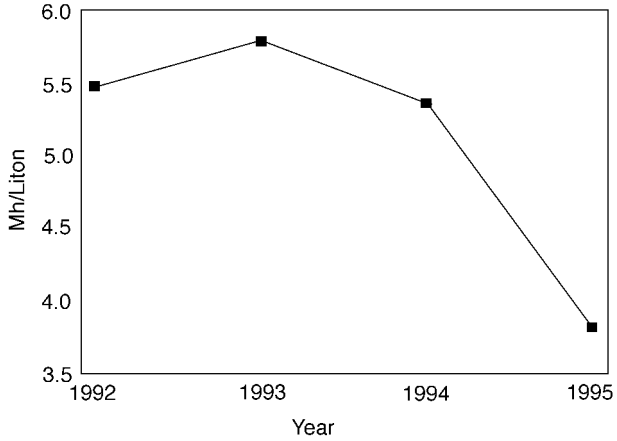


Fig. 6.2 HACCP benefits: increase in productivity in an ice cream plant.

- HACCP implementation has led to increased product safety. This is illustrated in Fig. 6.4, which shows the reduction in *Listeria* counts in ice creams after HACCP was introduced.
- HACCP implementation has reduced the frequencies of shut-down of process lines for cleaning and sanitation. Typically, shut-downs were needed once a day before HACCP, and were of the order of once a week after HACCP.
- HACCP implementation has allowed continuous manufacture for extended periods, without any stoppages due to process breakdown.
- The cost of poor quality product reduced by >10% within six months of HACCP implementation.

Because it involved a detailed and systematic analysis of production operations, and, in particular, a major overhaul of prerequisite systems, HACCP

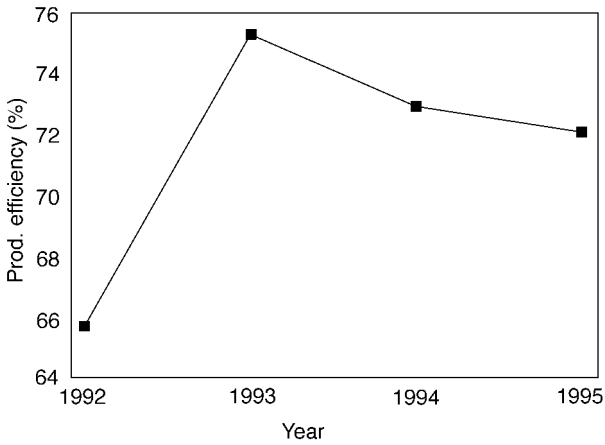


Fig. 6.3 HACCP benefits: increase in production efficiency.

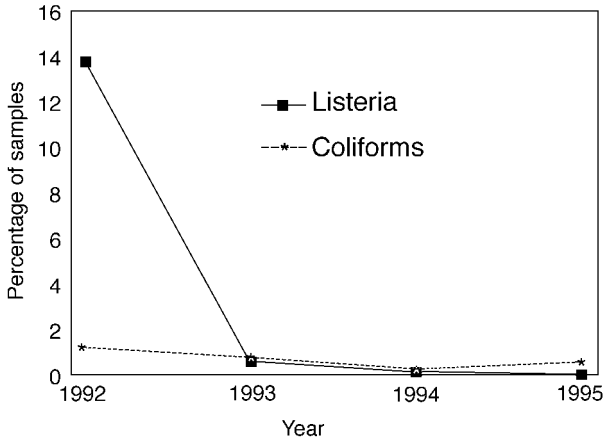


Fig. 6.4 HACCP benefits: product microbiology in an ice cream factory.

implementation has helped refocus manufacturing operations and led to several technical innovations in processing. More broadly, HACCP implementation has introduced a new culture of scientific/analytical thinking and problem solving in manufacture. Because it relies on the initiative of shop-floor staff in monitoring CCPs, HACCP implementation has also led to the empowerment of the workforce. HACCP has proved a self-sustaining system that encourages people to strive for continued improvement and innovation.

6.9 Difficulties in HACCP implementation

Some of the difficulties involved in HACCP implementation in the marine products sector as a whole include the following:

- Lack of information on hazards, especially pesticide and heavy metal residues and many areas of microbiological contamination.
- Poor control of raw material quality from suppliers. Landing and distribution of fish are unregulated activities. Raw material is obtained from vastly differing sources and there are significant qualitative differences. It has therefore proved very difficult to develop supplier HACCP systems.
- Lack of adequate expertise in and mechanisation of processes. The marine products industry has evolved as a largely manual operation due to a lack of technological development. This has led to reduced effectiveness in HACCP implementation, especially CCP monitoring.
- Lack of expertise in HACCP systems. This is especially true of regulatory authorities overseeing HACCP implementation. Thus, smaller companies have had difficulties in obtaining proper advice. HLL has been fortunate in drawing on internal expertise in food safety systems, both in India as well as through Unilever world-wide.

- Non-availability of flexible software support for HACCP documentation. Processing units in remote locations do not always have access to information technology, and software available for preparation of HACCP plans was limited. This has led to the creation of HACCP plans manually.

One of the key factors in ensuring successful HACCP implementation is an effective disease surveillance system. Information is constantly needed, for example, on outbreaks of disease, types and strains of pathogens involved and their behaviour, and types of foods that may be at risk. The USA has installed an 'early warning' system for tracking the occurrence of foodborne disease. This system, called the Foodborne Diseases Active Surveillance Network (Foodnet), tracks the incidence and sources of foodborne illness in the USA through active surveillance and other epidemiological studies, in collaboration with federal and state agencies.

Presently, most HACCP systems are designed using data on hazards available from developed countries such as the USA. However, the type of organisms relevant to them cannot necessarily be indicative of those present in the Indian context. This is particularly true given the low level of hygiene standards in India. Current national data cannot be relied on either as an accurate indicator of the prevalence of foodborne pathogens or their sources. As a result, one of the major weaknesses in the development of HACCP plans in India is hazard analysis. It is imperative that developing countries like India, which will develop an increasingly larger share of the global food market, create their own systems to track foodborne disease.

Such systems would also be greatly facilitated if data on microbial responses (growth, survival and death) from individual countries were collated in a more standardised form. Scientists everywhere are working to develop risk assessment models based on predictive microbiology, using data available through various experiments carried out in-house as well as from around the world. Indian science needs to be plugged into this network and actively collaborate.

6.10 The future of HACCP implementation in India

India requires dramatic changes in its existing food processing systems to improve food safety and reduce the incidence of foodborne illness associated with the consumption of processed and semi-processed products. Its food processing industry needs to adopt HACCP systems universally. Because food safety strategy is broadening to cover the entire farm-to-table continuum, HACCP implementation must also broaden to address all points along the farm-to-table chain. India not only needs to invest in food microbiology research to provide safe foods effectively; it is also important to prevent foodborne disease through minimum standards of hygiene across the supply chain, and through consumer and food handler education.

In the past, HACCP implementation in India has been held back by formidable infrastructural problems. The supply chain makes it difficult to establish even basic hygiene standards, and the food industry remains dominated by small-scale manufacturers with neither the resources nor the motivation to implement HACCP systems. Progress has also been held up by a cycle of poor knowledge of foodborne disease, resulting consumer indifference and inadequate legislation, and a consequent lack of incentive for manufacturers to invest in food safety. This cycle has only been broken in more lucrative export markets such as marine products and fresh fruit. Here a combination of external pressure from trading blocs such as the USA and the EU, and the growing value of the export trade, has encouraged both government and industry to work together and invest in HACCP systems. As a result, despite the formidable obstacles, both large businesses such as HLL and smaller manufacturers involved in the export trade have successfully implemented HACCP systems. However, the overall picture for the marine products industry as a whole remains patchy, and HACCP implementation in the rest of the food processing industry is still rare.

If HACCP implementation is to become more widespread, a key first step in India, as for other developing countries, is for the government to develop a national food safety strategy. As stated by the WHO, food safety has to be a shared responsibility between government, the consumer and industry. India will need to work toward such a national strategy to be able to compete globally. Indeed, a major force promoting HACCP in the long term may well be India's increasing involvement in international trade, exposing its consumers and its food industry to standards prevalent in the EU and the USA, for example.

The WHO provides guidelines for setting up and strengthening national food safety programmes. The four-step process is shown below:

Step 1: Assessment of food safety infrastructure and problems at national level: preparation of a country profile

Step 2: Preparation of a national food safety programme. This step includes actions such as:

- formulating a national food safety policy
- developing and/or updating food legislation
- strengthening food control systems
- promoting voluntary management systems for food safety assurance
- educating food handlers at all levels in food safety
- research and data collection

Step 3: Implementation of the national food safety programme

Step 4: Evaluation of food safety activities.

There are a number of steps that the Indian government can take in building a food safety programme and harnessing the support of consumers and the food industry. One is to work with industry in establishing common standards of GMP and GHP across the supply chain. These minimum standards then need to be reinforced by relevant legislation. Current food laws such as the PFA Act

need drastic revision and the creation of a food safety law will be a step in the right direction. Such laws need to foster a food safety culture among both consumers and industry, while at the same time not stifling technical innovation or imposing heavy costs on industry.

Other areas of legislation that need to be reassessed include laws on reservation of certain food categories for small-scale producers. A typical example is ice cream, which was, until recently, a reserved category that only small-scale manufacturers were allowed to manufacture. These types of products require high levels of safety which depend on investment in building an effective cold chain. Such investment is often beyond the small-scale manufacturer. This situation changed in 1999 and ice creams have now been de-reserved, allowing for increased investment by larger manufacturers. The government also needs to reform some aspects of commercial contract legislation. Currently, many aspects of contracts with farmers and other primary producers are not legally enforceable. As a result, while many multinational companies (MNCs) provide valuable support for primary producers, returns in terms of improvements in quality cannot be relied on. This gap in legislation inhibits the spread of improved safety standards back down the supply chain.

At the same time, government also needs to provide support and incentives for industry to implement HACCP systems. The model provided by the APEDA, for example, which provides fiscal incentives to implement HACCP systems, shows that such an approach can be successful in developing HACCP implementation among smaller manufacturers. All this will only be possible through focused training and awareness of food safety issues. Key requirements here are trained and committed personnel at all levels, including government itself; sufficient consumer awareness to make the safety of food an important issue in the marketplace; and the appropriate research infrastructure to identify the key food safety issues. If this can be achieved, the current cycle, which inhibits progress in countries such as India, can slowly but surely be reversed.

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